Checkpoint-Based Medical Agent — Full Pipeline

Complete Intake Form — All Fields

🧍 Demographics

- Full Name
- Age
- Date of Birth
- Sex (Male / Female / Other)
- Blood Group

📏 Physical Baseline

- Height
- Weight
- *(BMI auto-calculated)*

📍 Location

- Country
- City
- Region / Neighborhood
- Urban / Suburban / Rural

💼 Occupation

- Job type (desk / field / industrial / healthcare / student / unemployed)
- Chemical or dust exposure? (Y/N)
- Work hours per day

🏃 Lifestyle

- Smoker? (Never / Former / Current — how many/day)
- Alcohol? (Never / Occasional / Regular)
- Activity level (Sedentary / Light / Moderate / High)
- Sleep (hours per night)

- Diet type (Omnivore / Vegetarian / Vegan / High sugar / Other)

🧬 Medical History

- Pre-existing conditions (multi-select: Diabetes, Hypertension, Asthma, Thyroid, Heart disease, Other)
- Previous surgeries or hospitalizations
- Any chronic pain conditions

👪 Family History

- Diabetes (Y/N)
- Heart disease (Y/N)
- Cancer (Y/N)
- Mental illness (Y/N)
- Any other hereditary condition

💊 Current Medications

- Medication name + dosage (free text, repeatable)
- Any supplements or herbal medicines

⚠️ Allergies

- Drug allergies
- Food allergies
- Environmental allergies (pollen, dust, etc.)
- Reaction type (rash / anaphylaxis / breathing difficulty)

💉 Vaccinations

- Standard vaccines received (multi-select)
- Last tetanus shot
- Travel vaccines if any

🧠 Mental Health (optional but high signal)

- Stress level (Low / Medium / High)
- History of anxiety or depression (Y/N)
- Currently under psychiatric care (Y/N)

🌍 Environmental Exposure

- Near highway / factory / construction? (Y/N)
- Water source (municipal / well / bottled)
- Mold or dampness at home? (Y/N)
- Pets at home? (Y/N — allergy + zoonotic risk)

🌐 Travel

- Traveled outside home country in last 6 months? (Y/N)
- If yes — which regions?

Summary — Field Count

Section	Fields
Demographics	5
Physical	2
Location	4
Occupation	3
Lifestyle	5
Medical history	3
Family history	5
Medications	2
Allergies	4
Vaccinations	3
Mental health	3
Environment	5
Travel	2
Total	**~46 fields**

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46 fields sounds like a lot but most are **dropdowns or Y/N** — realistically **5–7 minutes** to fill. This gives any doctor or AI agent a near-complete prior before the first symptom is even described.

STAGE 1 — Form Intake

Checkpoint: Form fully submitted and validated

- Collect all ~46 fields
- Validate completeness (flag missing critical fields)
- Store as **User Profile Object**

```
UserProfile {
  demographics, vitals, location,
  lifestyle, medical_history, family_history,
  medications, allergies, vaccinations,
  environment, travel
}
```

✅ Checkpoint clears when: form is submitted with no critical fields empty

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STAGE 2 — Regional Research (Web Search)

Checkpoint: Risk profile built for user's geography

Agent runs a background search:

- *"Common diseases in [city/region], [country]"*
- *"Endemic diseases [region] [current season]"*
- *"Air quality / water quality [city]"*
- *"Disease outbreaks [region] last 6 months"*

Output: Regional Risk Tag List

```
RegionalRisk {
  endemic: [Dengue, Typhoid, ...],
  seasonal: [Flu, Pollen allergy, ...],
  environmental: [High PM2.5, Hard water, ...],
  active_outbreaks: [...]
}
```

✅ Checkpoint clears when: regional risk list is generated

STAGE 3 — Preprocessing (Risk Fusion)

Checkpoint: Unified Prior Risk Profile ready

Cross-merge UserProfile + RegionalRisk:

```
PriorRiskProfile {

  high_risk_conditions: [
    "Diabetic → watch for infections, neuropathy"
    "Smoker in high PM2.5 city → respiratory priority"
    "Unvaccinated + Dengue region → flag fever hard"
  ],

  ruled_out: [
    "Vaccinated for Hep B → deprioritize"
    "No travel → rule out exotic disease"
  ],

  watchlist: [
    // top 5-7 most probable conditions
    // ranked by profile + region match
```

```
  ]
}
```

This becomes the **agent's invisible prior** — it never shows this to the user but every question is now guided by it.

✅ Checkpoint clears when: watchlist is ranked and ready

STAGE 4 — Symptom Questioning

Checkpoint: Full symptom picture captured

Agent now talks to the user. **One question at a time.**

Block A — Chief Complaint

> "What is bothering you the most right now?"

Block B — Duration (mandatory)

> "When did this start?"
> "Has it been constant or does it come and go?"
> "Is it getting worse, staying the same, or improving?"

Duration logic:

Duration	Implication
< 3 days	Likely acute / infectious
3–14 days	Sub-acute, monitor closely
> 2 weeks	Chronic / needs investigation
Recurring	Pattern-based or autoimmune

Block C — SOCRATES drill
Site → Onset → Character → Radiation → Associated symptoms → Timing → Exacerbating/Relieving → Severity (1–10)

Block D — Red Flag Screen
Hard check — if any yes → immediately escalate, stop pipeline

Block E — Functional Impact

- Eating / sleeping / working normally?
- Took any medication for it?

✅ Checkpoint clears when: all blocks A–E completed, no red flags

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STAGE 5 — Personal Data Cross-Reference

Checkpoint: Most probable cause identified

Agent now silently reasons:

```
SymptomSet (from Stage 4)
    +
PriorRiskProfile (from Stage 3)
    ↓
Differential Diagnosis List
```

Matching logic:

```
For each condition in watchlist:
  score += symptom match weight
  score += demographic match (age, sex, BMI)
  score += lifestyle match (smoker, sedentary)
  score += family history match
  score += regional risk match
  score += medication side effect match

→ Rank by score
→ Top 3 = probable causes
```

If no strong match found:

```
IF max_score < threshold:
  flag = "GENERIC or SEVERE"
  → go to Stage 6 branching
```

✅ Checkpoint clears when: differential ranked OR generic/severe flag raised

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STAGE 6 — Final Classification → Diagnosis Pathway

```
                    ┌──────────────────┐
                    │  Probable match  │
                    │    found?        │
                    └──────────────────┘
                              │
                              │
              ┌───────────────────────────┐
             YES                          NO
              │                           │
    ┌─────────▼──────────┐      ┌──────────▼──────────┐
    │ Severity Check     │      │  Generic vs Severe  │
    │ (functional impact │      │  branch             │
    │ + red flag history)│      └─────────────────────┘
    └────────────────────┘                │
              │                           │
    ┌─────────▼──────┐          ┌──────────┴──────────┐
    │                │         GENERIC             SEVERE
    │                │            │                   │
  HIGH            LOW/MED         │                   │
    │                │       Viral fever,         Could be rare,
  See doctor     Home care   common cold,         autoimmune,
  within 24hrs   + monitor   gastro, fatigue      neurological,
                             → Rest + OTC          cancer signal
                                                  → ER / specialist
```

Final Output to User

```
🟡 MODERATE SEVERITY

Most probable cause: Viral Upper Respiratory Infection
Confidence: High (symptom match + flu season in your region)

Based on:
→ Your symptoms (sore throat, fatigue, 3 days duration)
→ Your location (Mumbai, current flu season)
→ Your profile (no chronic conditions, vaccinated)

Recommended action: Rest, hydration, OTC antihistamine.
If fever exceeds 103°F or persists beyond 5 days → see a doctor.

⚠ Ruled out but watch for:
→ Dengue (endemic in your region) — monitor for rash or joint pain
```

Pipeline Summary

```
Stage 1  →  Form filled           →  UserProfile
Stage 2  →  Web research          →  RegionalRisk
Stage 3  →  Fusion                →  PriorRiskProfile (invisible prior)
Stage 4  →  Symptom questioning   →  SymptomSet
Stage 5  →  Cross-reference       →  Differential ranked / flag raised
Stage 6  →  Classification        →  Output + action recommendation
```

Each stage only proceeds when its checkpoint is cleared — **no skipping, no assumptions carried forward without data.**